

M3: Mastering Jetpack Compose Layouts: Building Declarative UIs with Composable Functions and Layouts

Tuesday, May 26, 2026 15:35

Overview

Introduction

Hello and welcome to Module 3 of COMP-304. In this module, you will learn about Mastering Jetpack Compose Layouts: Building Declarative UIs with Composable Functions and Layouts.

Learning Outcomes

At the end of this module, you will be able to:

1. Discuss the fundamental principles of Jetpack Compose and the differences between declarative and imperative UIs.
2. Identify and explain the purpose and usage of composable functions in Jetpack Compose.
3. Develop and implement UI components using composable functions, previews, and modifiers.
4. Design and construct complex UI layouts using Column, Row, Box, and Lists in Jetpack Compose.
5. Evaluate the efficiency and effectiveness of different layout structures in Jetpack Compose.
6. Apply Jetpack Compose best practices to create responsive and maintainable user interfaces

Checklist

There are several tasks you'll need to complete:

1. Review the online content for this module, which includes reviewing the module's key terms and concepts
2. Run all class examples

Key Terms and Concepts

Here are a few terms you need to be familiar with.

Term	Definition
Jetpack Compose	A modern UI toolkit introduced by Google for creating less code-intensive and more intuitive UIs in Android using a declarative approach.
Composable Functions	Fundamental building blocks in Jetpack Compose, marked with the @Composable annotation, used to define parts of the UI.
Imperative UI Programming	Involves manual handling of UI changes and state, requiring explicit updates to the UI.
Declarative UI Programming	Focuses on defining the UI as a function of state, where the framework handles updates automatically. Its life cycle methods are similar to activity life cycle methods.
Modifiers	Used in Jetpack Compose to configure and customize the layout and appearance of UI components.
Column Layout	Arranges child composables vertically in Jetpack Compose.
Row Layout	Arranges child composables horizontally in Jetpack Compose.
Box Layout	Layers components on top of each other, allowing for overlapping UI elements
LazyColumn	A component in Jetpack Compose for displaying scrollable lists vertically, rendering only visible items to optimize performance.
LazyRow	Similar to LazyColumn but arranges items horizontally.

Weekly Topic - Compose layout basics:

Summary

Key Take-Aways

Here are a few key take-aways from this module.

Jetpack Compose makes it much easier to design and build your app's UI. Compose transforms state into UI elements, via:

1. Composition of elements
2. Layout of elements
3. Drawing of elements

•

Don't forget to complete the items on your checklist.

Summary Checklist

Don't forget to complete the items on your checklist:

1. Review the online content for this module, which includes reviewing the module's key terms and concepts.
2. Test all the class examples.
3. Read articles/tutorials posted in Learning Activities.

Works Cited

- Reference book
- Android Documentation: <https://developer.android.com/develop/ui/compose/layouts/basics>

Coming up next, we'll be looking at the weekly topic for module 4 as identified in the approved course outline.

COMP-304: Mobile Apps Development

